

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – CONCEPT MAPPING
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO2 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	Kamalesh T	Reg. No.:	192424228
Section / Batch:	2024	Date:	31/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
CONCEPT MAPPING	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1	100
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Q1. Consider the concept map: Construct the map showing how a recurrence expands into a tree, how each level contributes a cost, and how summing across levels yields total complexity. Justify how the hierarchical relationships in the map make the recursion tree method an effective analysis tool.

Code of Conduct

I, P. Kamalesh T(192424228) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kamalesh T



RUBRICS FOR CALCULATION

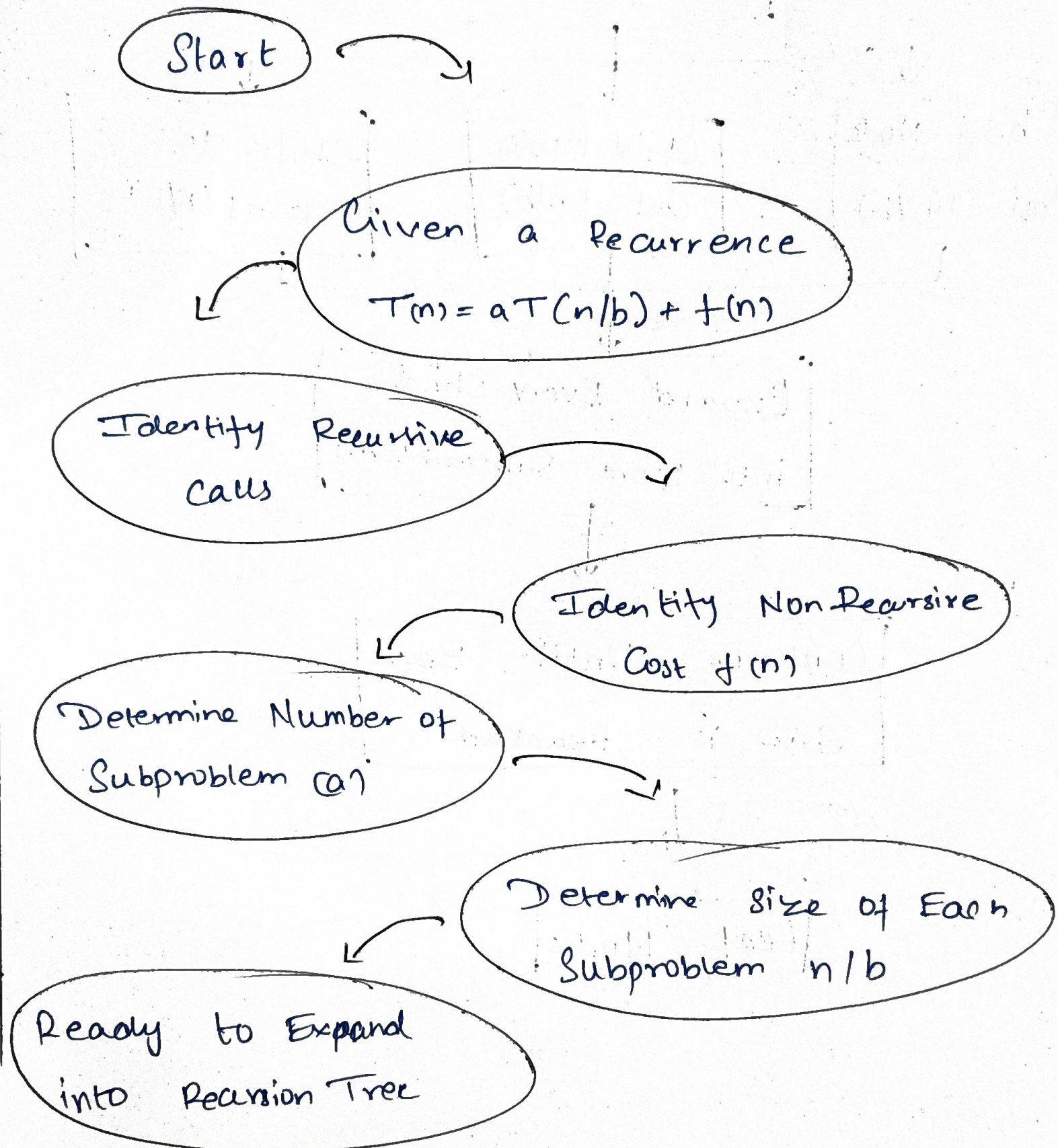
Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

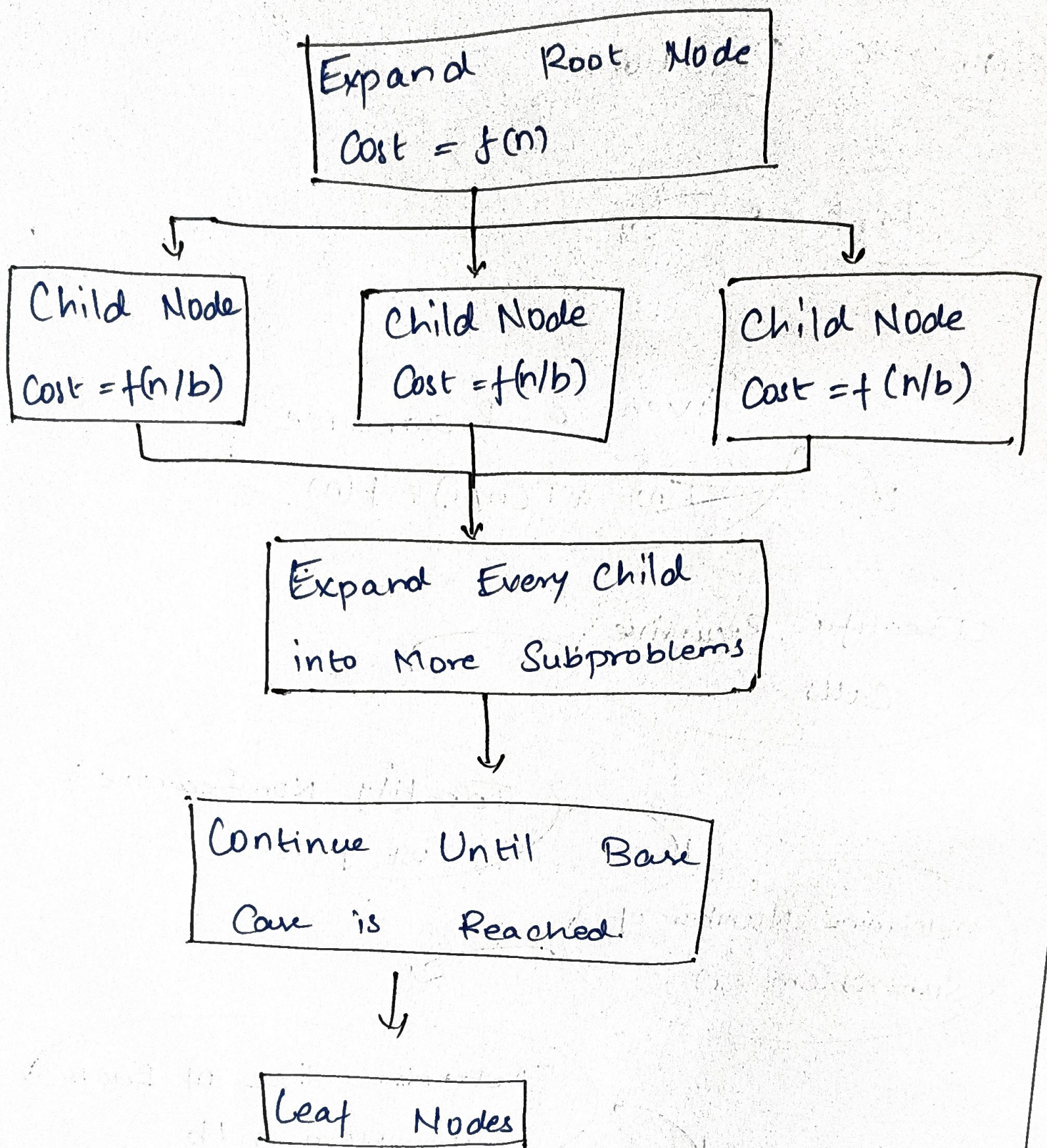
Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (100 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Concept Mapping.

Recurrence Relation:

Tree Expansion



Total Time Complexity

Sum Cost of All Levels



Total Cost of Tree



Simplify Mathematical Expression



Determine Asymptotic Time Complexity



Example Results
 $O(n \log n)$, $O(n^2)$, $O(n)$



Interpret Efficiency of Recursive Algorithm.

Cost Calculation

Calculate Root
Cost $f(n)$

Calculate Cost
of Level 1

Calculate Cost
of Level 2

Continue for
Every Level

Sum Cost of Nodes
at Each Level

Obtain Total Level
Cost